SETK 4643
DOWNSTREAM BIOPROCESSING

Disruption of Cell

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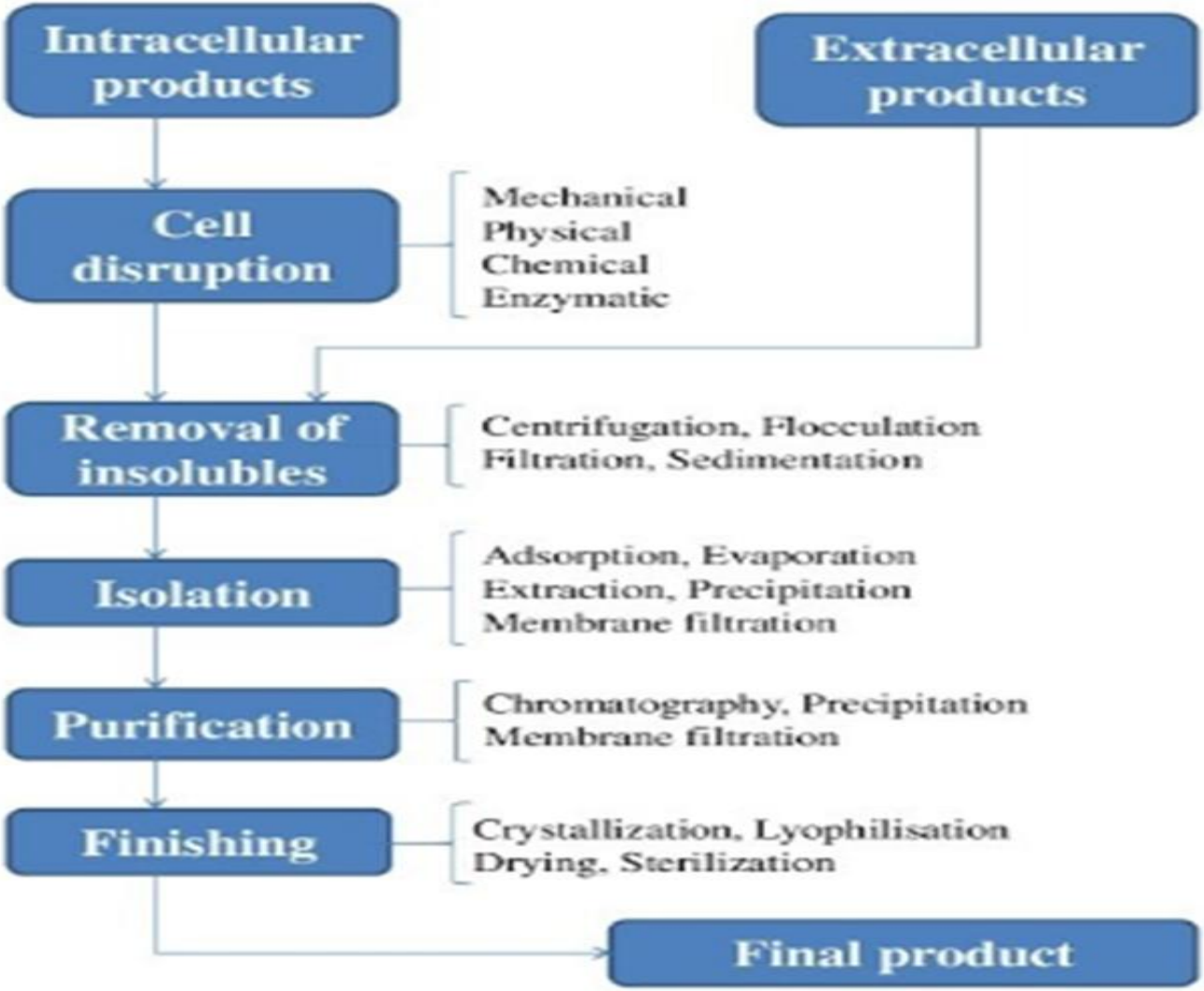
Downstream process



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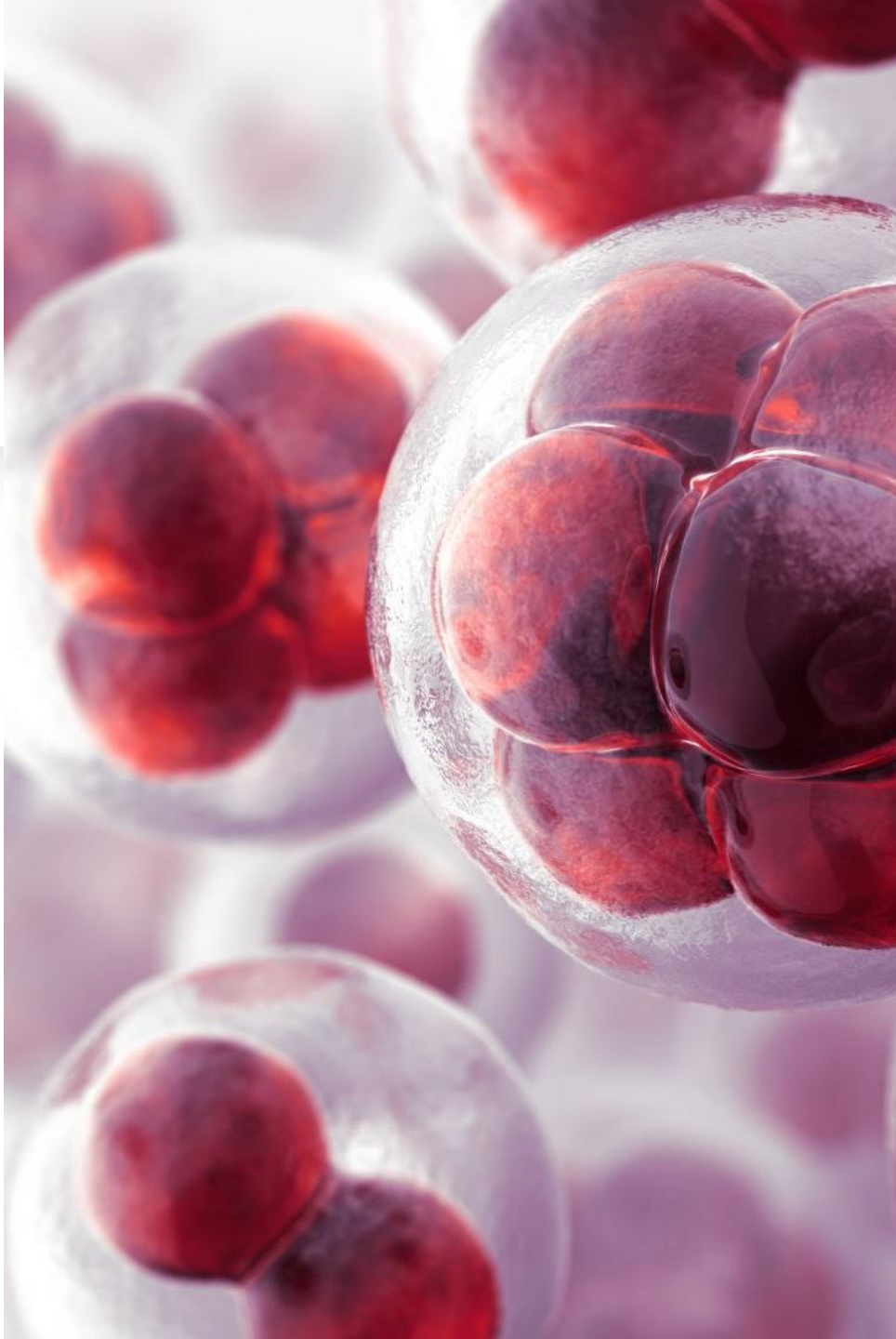
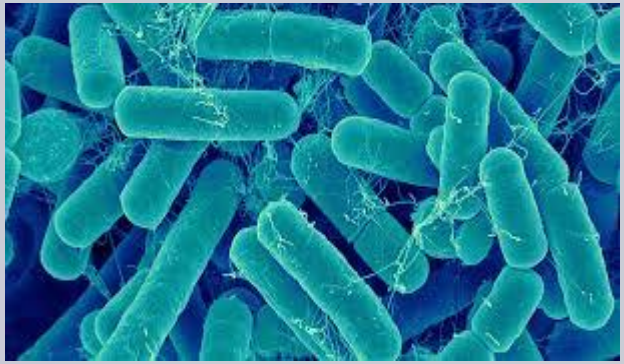
Learning outcome

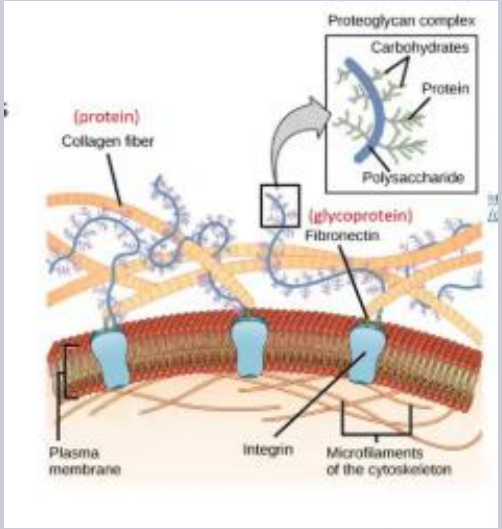
- Explain the concepts of cell disruption
- Identify different types of biological cells
- Identify, Explain, and Differentiate between the mechanical and non-mechanical methods for cell disruption

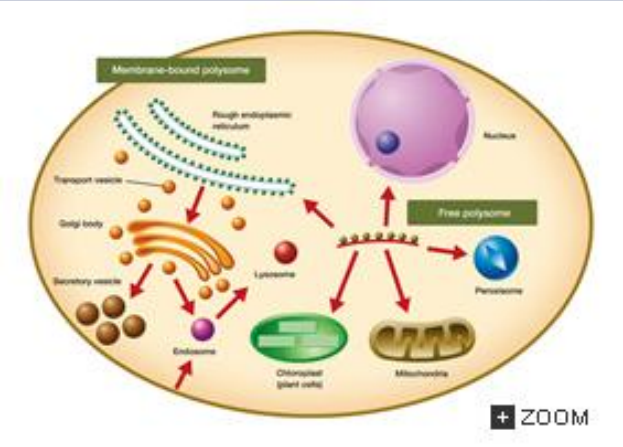


INTRODUCTION

Biological products synthesized by fermentation or cell culture are either intracellular or extracellular.







Cell itself

- Bakers yeast

Extracellular type

- Antibiotics
- Alcohol

Intracellular type

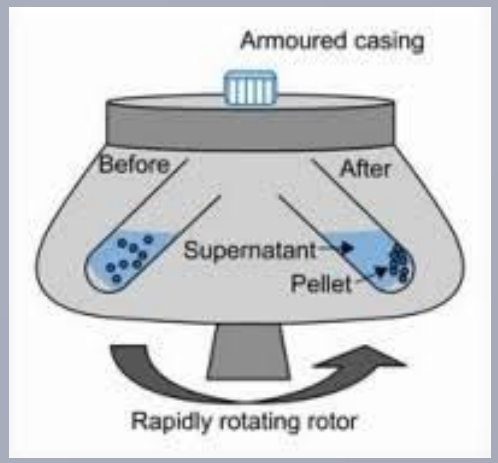
-Soluble form or as inclusion bodies (fine particles)

- Enzymes
- Vitamins

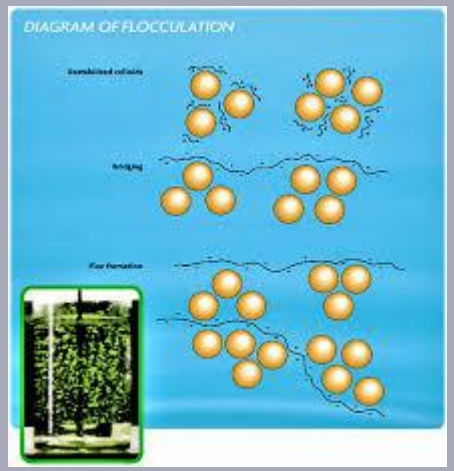
How to recover if the product is the cells itself?

- Separation of particles

Filtration



Centrifugation



Flocculation

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How to recover if the product is excreted outside the cell??

Separation of particle

- Centrifugation or filtration
- Flocculation

Extraction

- Liquid-liquid extraction

Concentration

- Chromatographic technique
- Crystallization
- Adsorption

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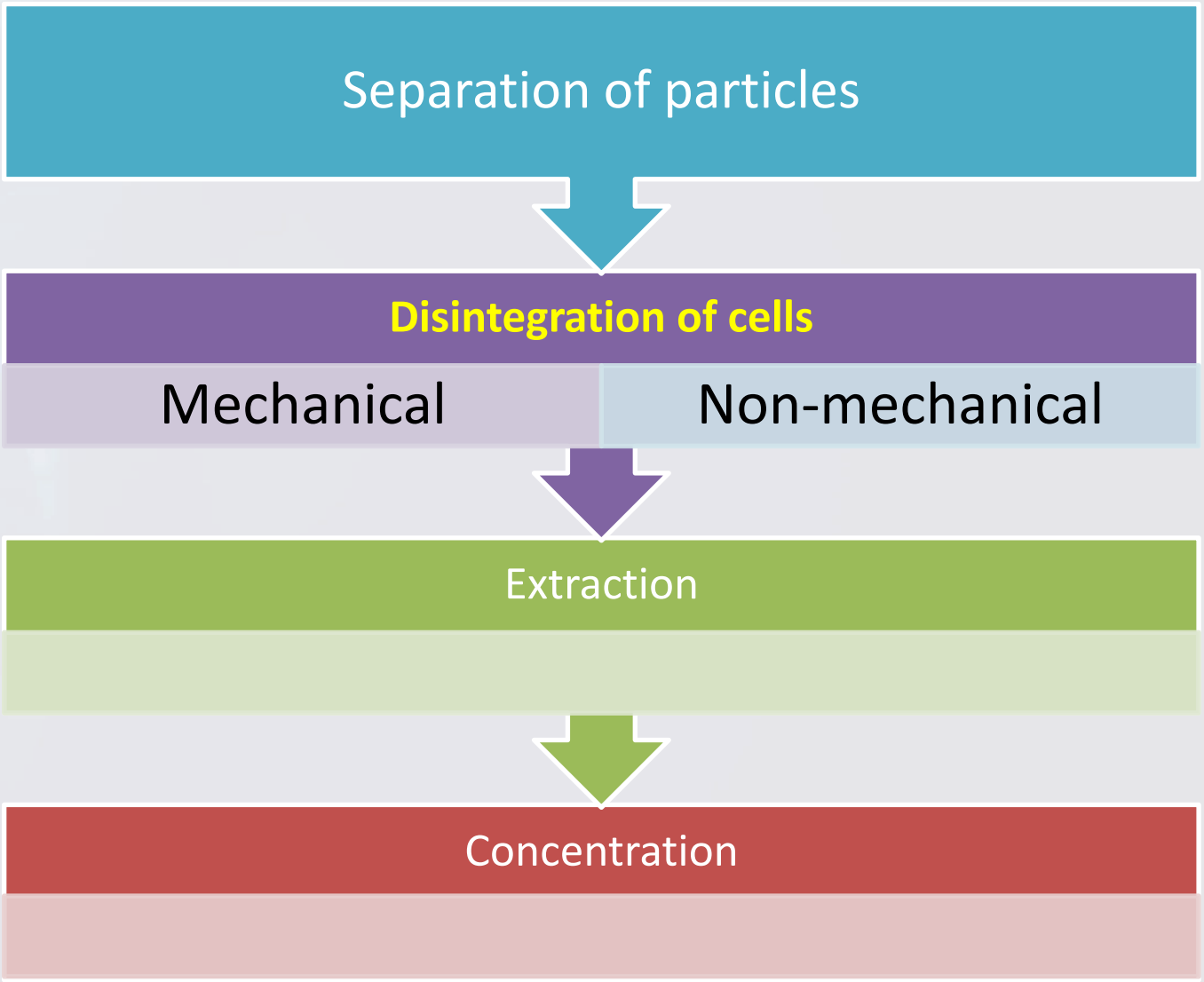
How to recover if the product is intracellular?



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Video time

https://www.google.com/search?q=cell+disruption+youtube&ei=Kq5sY9WABPPmz7sPs-qRWA#fpstate=ive&vld=cid:a2a3c96b,vid:DFe-AyB0vc8

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CELL DISRUPTION

- Cell disruption/cell rupture/ lysis
- A large no of recombinant products–protein, or DNA accumulate in larger quantities within cells (intracellular)
- Cells need to be disrupted first to release products into the liquid medium
- The process of breaking cell walls and/or membranes to release intracellular products containing molecules or particles of interest.
- Some products must first be extracted from biological tissues (e.g., insulin historically obtained from pig or cow pancreas as a tissue-derived substance). In such cases, the tissue is typically homogenized or ground into a cell suspension before undergoing cell disruption to release the desired product.
- Nowadays, insulin is primarily produced using recombinant DNA technology. This process involves genetically modifying microorganisms, such as Escherichia coli or Saccharomyces cerevisiae, to produce human insulin. The gene encoding human insulin is inserted into these host cells, which then synthesize the insulin protein during their growth.

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What is considered effective cell disruption??

- High yield of target product
- Low-cost
- Low energy consumption
- Lessening waste or by-products (non-targeted products)
- Limiting damage or degradation of products due to high temperature or excessive force

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Cells

- Different types of cells need to be disrupted in the bio-industry:
    1) Gram-positive bacterial cells
    2) Gram-negative bacterial cells
    3) Yeast and Mold cell
    4) Cultured mammalian cells
    5) Cultured plant cells
    6) Algae cells- microalgae and macroalgae

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Parameter	Gram-positive bacteria	Gram-negative bacteria
Cell Wall	A single-layered, smooth cell wall	A double-layered, wavy cell-wall
Cell Wall thickness	The thickness of the cell wall is 20 to 80 nanometres	The thickness of the cell wall is 8 to 10 nanometres
Peptidoglycan Layer	It is a thick layer/ also can be multi-layered.	It is a thin layer/ often single-layered.
Teichoic acids	Teichoic acids are present.	Teichoic acids are not present.
Lipopolysaccharide	Lipopolysaccharide is not present.	Lipopolysaccharide is present.
Outer membrane	The outer membrane is not present.	The outer membrane is mostly present.
Lipid content	The Lipid content is very low.	The Lipid content is 20% to 30%.
Resistance to Antibiotic	These are very susceptible to antibiotics.	These are very resistant to antibiotics.

Biology Lectures

Gram positive	Count (n)	Gram negative	Count (n)	Fungus/Yeast	Count (n)
Aspergillus niger	1	*Chromobacterium violaceum*	1	*Candida albicans*	8
Bacillus cereus	1	*Escherichia coli*	14	*Candida dibliniensis*	1
Bacillus subtilis	5	*Klebsiella pneumonia*	5	*Candida glabrata*	1
Enterococcus durans	1	*Proteus mirabilis*	2	*Candida tropicalis*	1
Enterococcus faecalis	4	*Proteus vulgaris*	1	*Pichia membranifaciens*	1
Enterococcus faecium	1	*Pseudomonas aeruginosa*	9	*Saccharomyces cerevisiae*	3
Listeria monocytogenes	3	*Pseudomonas fluorescens*	1	*Schizosaccharomyces pombe*	1
Micrococcus luteus	2	*Salmonella enteritidis*	3	*Zygosaccharomyces rouxii*	1
Methicillin Resistant Staphylococcus aureus (MRSA)	5	*Shigella flexneri*	2		
Staphylococcus aureus	14	*Salmonella infantis*	1		

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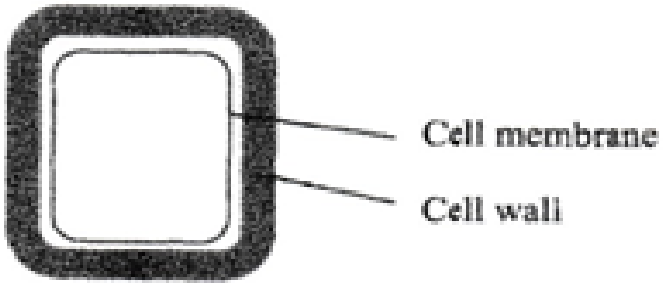
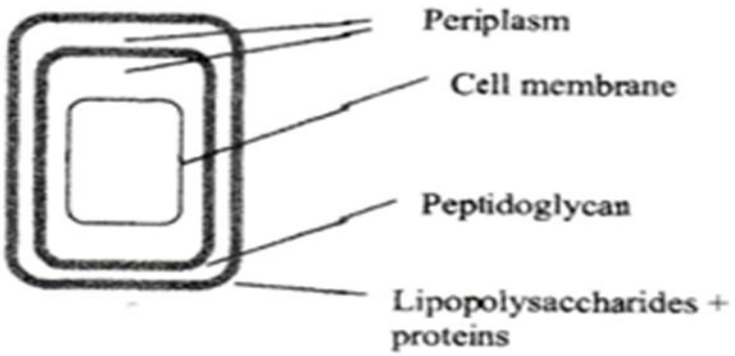
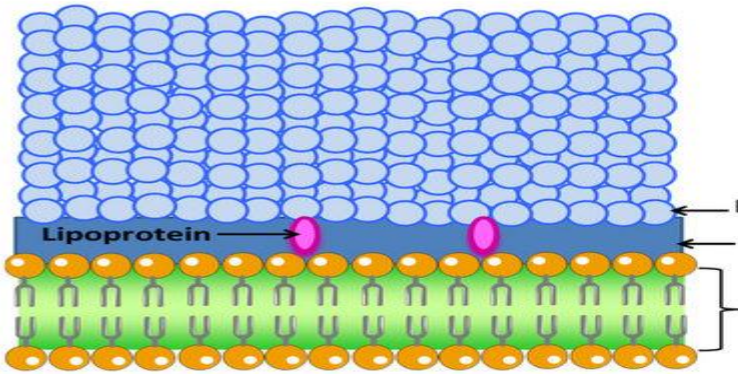
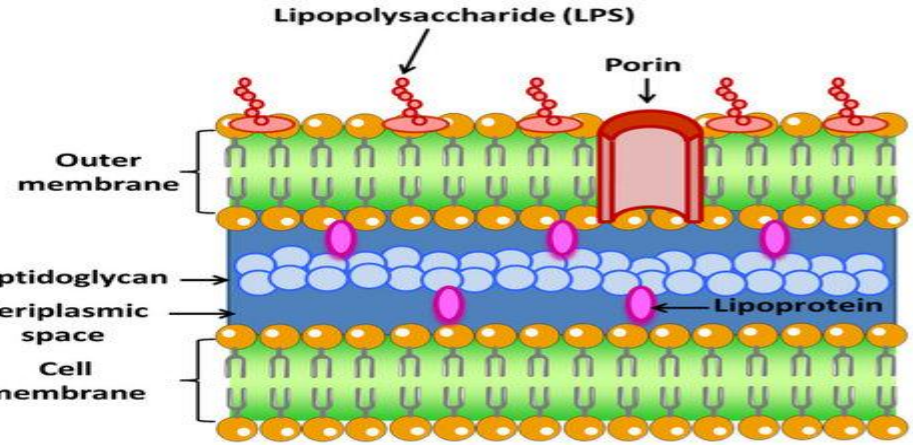






Gram positive



Gram negative

- ✓ Main barrier-cell wall (0.02 to 0.04 microns)
- ✓ composed of thick peptidoglycan, teichoic acid, and polysaccharides
- ✓ the plasma or cell membrane is made up of phospholipids and proteins (fragile).
- ✓ susceptible to lysis by lysozyme

- ✓ Do not have a distinct cell wall - multilayered envelops.
- ✓ The peptidoglycan layer - thinner than in gram positive bacteria.
- ✓ An external layer - lipopolysaccharides and proteins.
- ✓ Presence of the periplasm layer - large number of proteins, particularly recombinant proteins.
- ✓ Osmotic shock can be used to recover periplasmic (the region between plasma membrane and outer membrane)

Protocol

Large-Scale Extraction of Recombinant Proteins from Bacteria

Richard J. Simpson

INTRODUCTION

Bacteria are particularly convenient for producing recombinant proteins for purification purposes. Suitable extraction methods for bacterial cells include sonication, glass bead milling, grinding with alumina or sand, high-pressure shearing using the French pressure cell (French Press), and lysozyme treatment. These procedures are applicable for preparing extracts from a variety of Gram-negative bacteria such as *Escherichia coli* and *Klebsiella pneumoniae*, and Gram-positive bacteria such as *Bacillus subtilis*. Disruption of bacterial cells by enzymatic means is commonly used because a relatively uniform treatment is obtained when cells are in suspension. This protocol describes the enzymatic disruption of *E. coli*.

The method used to disrupt the bacteria cell- bead milling, French pressure cell, lysozyme, osmotic shock,

Simpson R. J. (2010). Large-scale extraction of recombinant proteins from bacteria. Cold Spring Harbor protocols, 2010(9), pdb.prot5484. https://doi.org/10.1101/pdb.prot5484

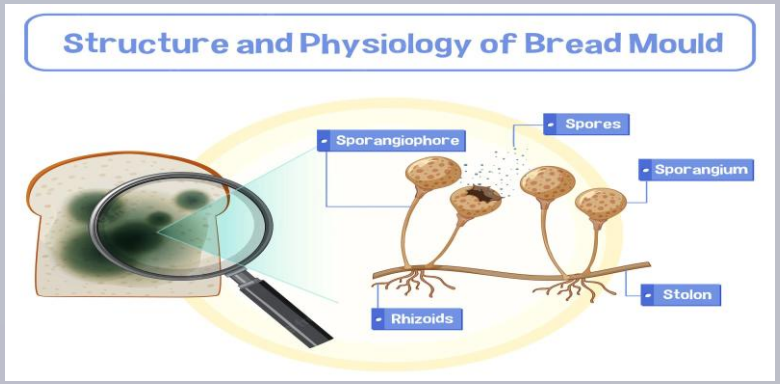
Yeast and Mold

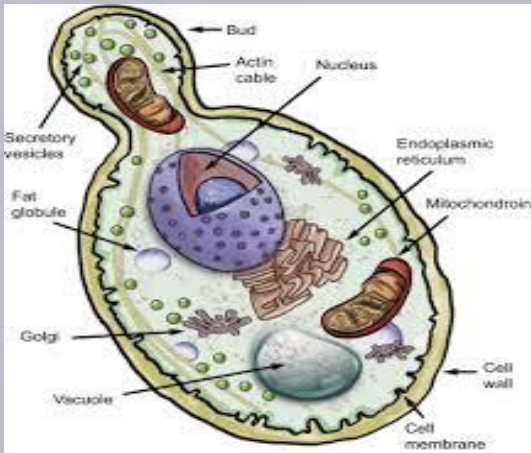
Yeast

- ✓ Unicellular have thick cell walls (0.1 to 0.2 microns)
- ✓ Composed of **polysaccharides** such as glucans, mannans and chitins.
- ✓ The plasma membrane is composed of phospholipids and lipoproteins.
- ✓ Yeast is a proven host for the production of **recombinant proteins.**
- ✓ cell disruption techniques for recovery of intracellular bio-active compounds from Baker's yeast including mechanical (bead mill, high pressure homogenization, ultrasonication), and non-mechanical (electrical, physical, chemical, and enzymatic) techniques

Molds

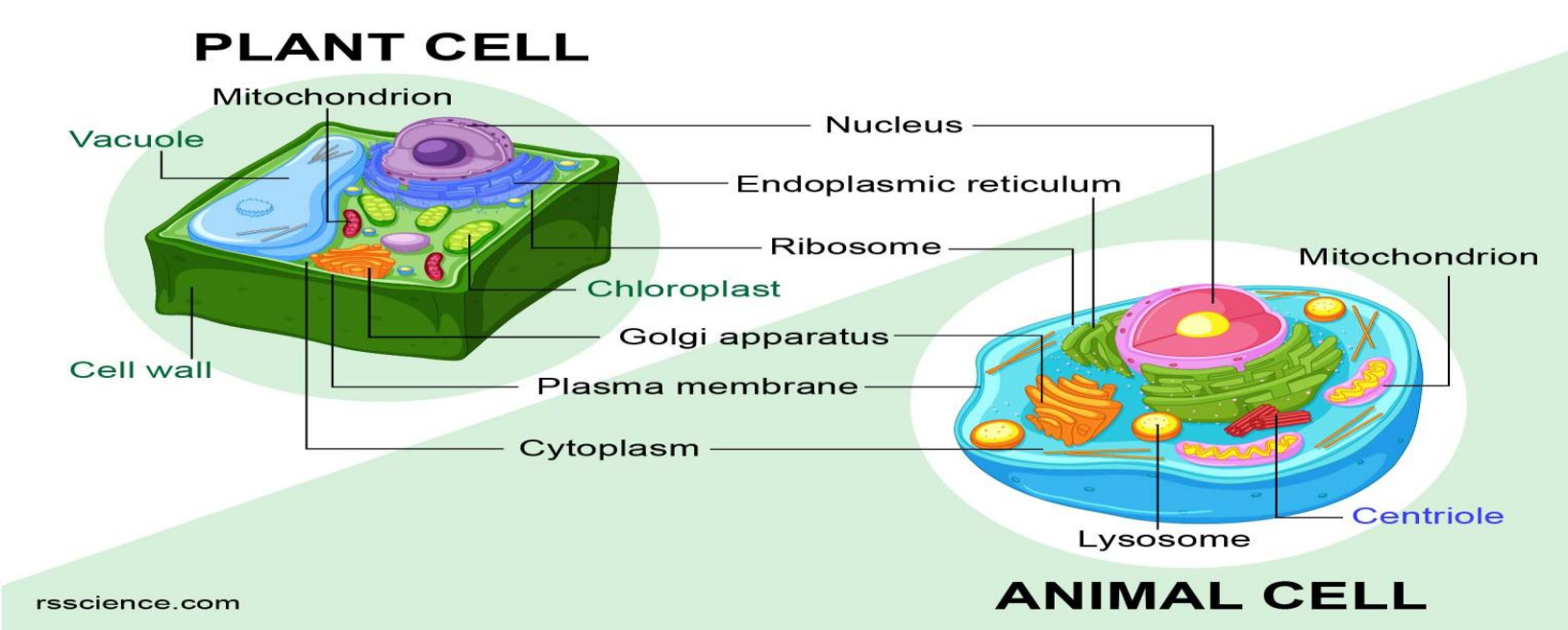
- ✓ are largely similar to yeast cells in terms of the cell wall and plasma membrane
- ✓ Bigger and Multicellular





Mammalian and Plant Cells

- Mammalian cells do not possess the cell wall and are quite fragile.
- Plant cells have very thick cell walls mainly composed of cellulose and other polysaccharides.
- Plant cells are difficult to disrupt
- Cultured plant cells are less robust than real plant cells



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Cells Distruption

- **Cell wall and cell membrane needs to be disrupted to recover intracellular products.**

Physical methods

- Disruption in bead mill
- Disruption using a rotor-stator mill
- Disruption using French press
- Disruption using ultrasonic vibrations

Chemical and physicochemical methods

- Disruption using detergents
- Disruption using enzymes e.g. lysozyme
- Disruption using solvents
- Disruption using osmotic shock

Emerging / Advanced Methods

- Pulsed electric field (PEF)
- Microwave-assisted disruption
- Supercritical CO_2 extraction

Physical methods mainly cause mechanical disruption of both the cell wall and membrane, while chemical and physicochemical methods act by weakening or solubilizing cell structures.

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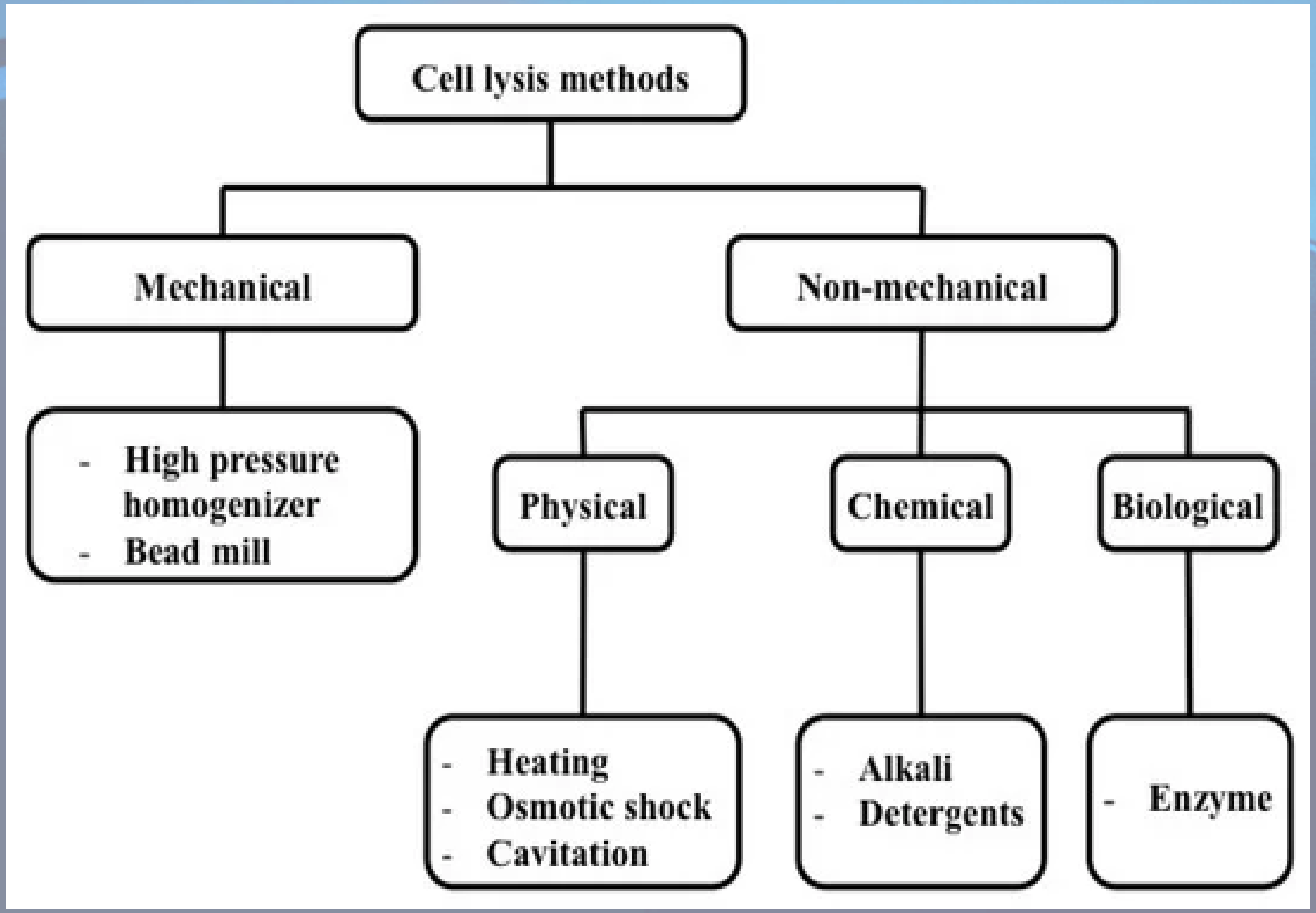
Cell lysis methods
Mechanical
Non-mechanical
- High pressure homogenizer
- Bead mill
Physical
Chemical
Biological
- Heating
- Osmotic shock
- Cavitation
- Alkali
- Detergents
- Enzyme

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Discussion time (Jamboard)

- Describe the method used to disrupt the cell- (Group1- bead milling, ultrasonic vibration, Group 2-French pressure cell (high pressure homogenization), Group 3- lysozyme, Group 4-osmotic shock, Group 5-detergent, and organic solvent
- List out the advantages and disadvantages

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Physical methods

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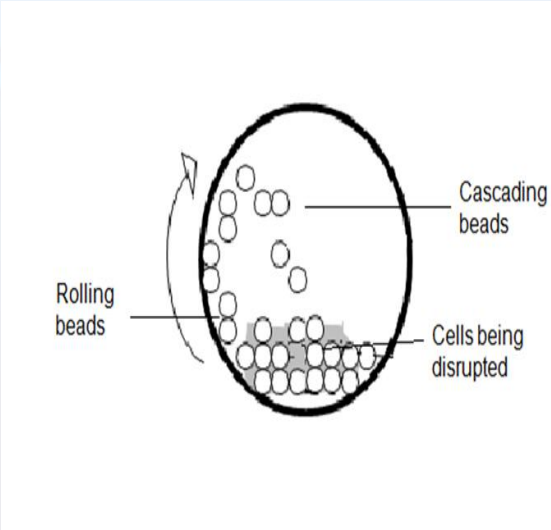
Bead mill



- Tubular vessel made of metal or thick glass
- Cell suspension is placed along with small metal or glass beads.
- The tubular vessel - rotated about its axis - beads start rolling away from the direction of the vessel rotation.
- Disruption takes place due to the grinding action of the rolling beads and the impact resulting from the cascading ones
- Disadvantage-Bead milling can generate enormous amounts of heat
- Cryogenic bead milling : Liquid nitrogen or glycol cooled unit
- Application: Yeast, animal and plant tissue
- Small scale: Few kilograms of yeast cells per hour
- Large scale: Hundreds of kilograms per hour.

https://www.youtube.com/watch?v=P_CPU5MDHvs

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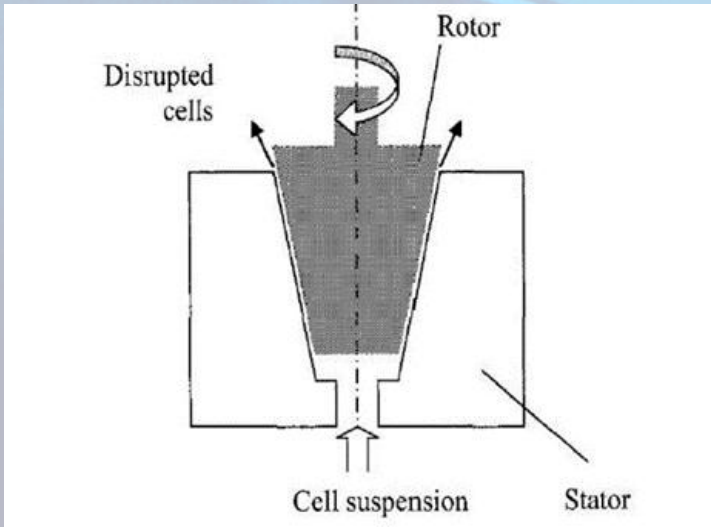
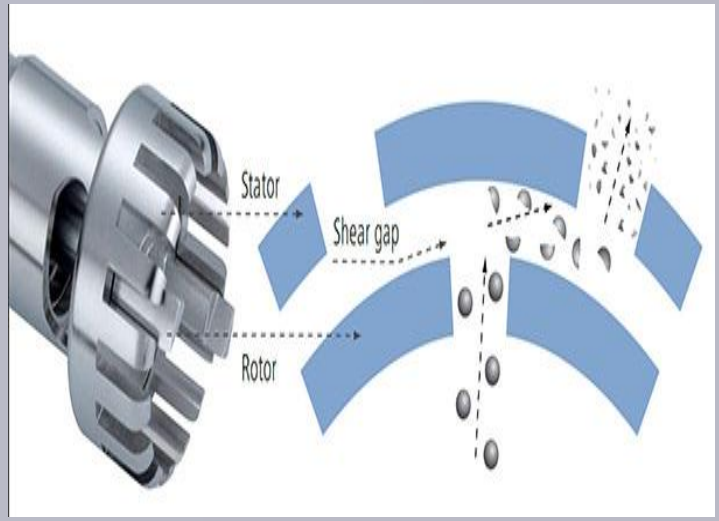
Rotor-Stator Mill

- This device consists of stationary block called stator and truncated cone rotating object called the rotor.
- Rotation speed: 10,000 to 50,000 rpm.
- Mechanism of cell disruption: High shear and turbulence
- The cell suspension is fed into the tiny gap between the rotating rotor and the fixed stator.
- Application: plant and animal tissue.
- Operation: Single and multi-pass mode (material is sent back into the device for more complete disruption).
- https://www.youtube.com/watch?v=sAFEtrWs7dk&t=17s





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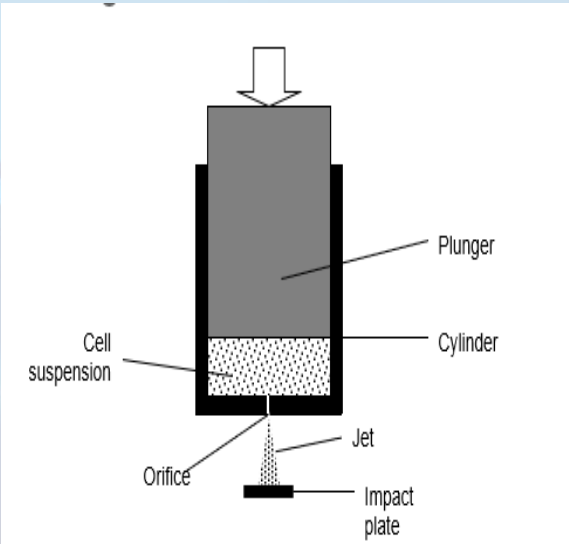
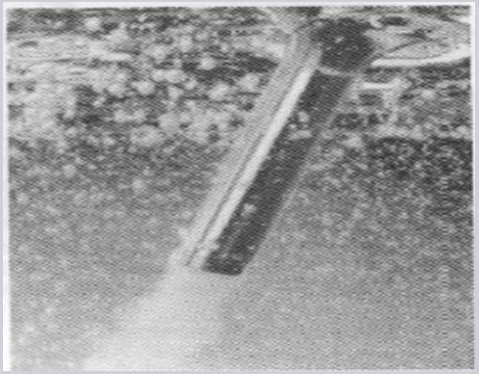
French Press

- A cylinder fitted with a plunger which is connected to a hydraulic press.
- The cylinder is provided with an orifice (suspension emerges at very high velocity) and impact plate (jet impinges causing cell disruption).
- Mechanism of cell disruption: High shear rates influence by the cells within the orifice.
- Typical volumes: few milliliter to hundred milliliter
- Typical operating pressure: 10,000 to 50,000 psig.
- Application: Intracellular proteins and DNA from bacterial and plant cells



Ultrasonic Vibration

- The cells are subjected to ultrasonic vibration by introducing an ultrasonic vibration emitting tip into the cell suspension.
- The ultrasonic vibration could be emitted continuously or in the form of short pulses.
- A frequency of 25 kHz is commonly used for cell disruption.
- These high frequency vibration cause cavitation where the formation of tiny bubbles within the liquid medium.
- When these bubbles reach resonance size, they collapse releasing mechanical energy in the form of shock waves and travel through the liquid, resulting in a shear force that will disrupt cells.



Cavitation during ultrasonic cell disruption. The shock waves disrupt cell present in suspension.

https://www.youtube.com/watch?v=GeHVIspOcig

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Ultrasonic Vibration

- The ultrasonic energy transmitted to the liquid through a sonicator or tip or horn.
- The rapid vibration generates a large amount of heat that must be dissipated by turning the sonicator off every few minutes.
- Application: Bacterial cells (E. coli; 30 to 60 seconds), yeast cell (2 to 10 minutes).
- Ultrasonic vibration is frequently used in conjunction with chemical (detergents) cell disruption methods.
- This approach could reduce the amount of energy needed for disruption.

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Example

A batch of yeast cells was disrupted using ultrasonic vibrations to release an intracellular product. The concentration of released product in the solution was measured during the process :

TIME (S)	CONCENTRATION (mg/ml)
60	3.49
120	4.56

If the ultrasonic cell disruption were carried out for 240 second, predict the product concentration

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Based on equation;

$$\frac{C}{C_{\max}} = 1 - \exp\left(-\frac{t}{\theta}\right)$$

$$3.49 = C_{max}\left(1 - \exp\left(-\frac{60}{\theta}\right)\right)$$

$$4.56 = C_{max}\left(1 - \exp\left(-\frac{120}{\theta}\right)\right)$$

Solving these equations simultaneously we get:

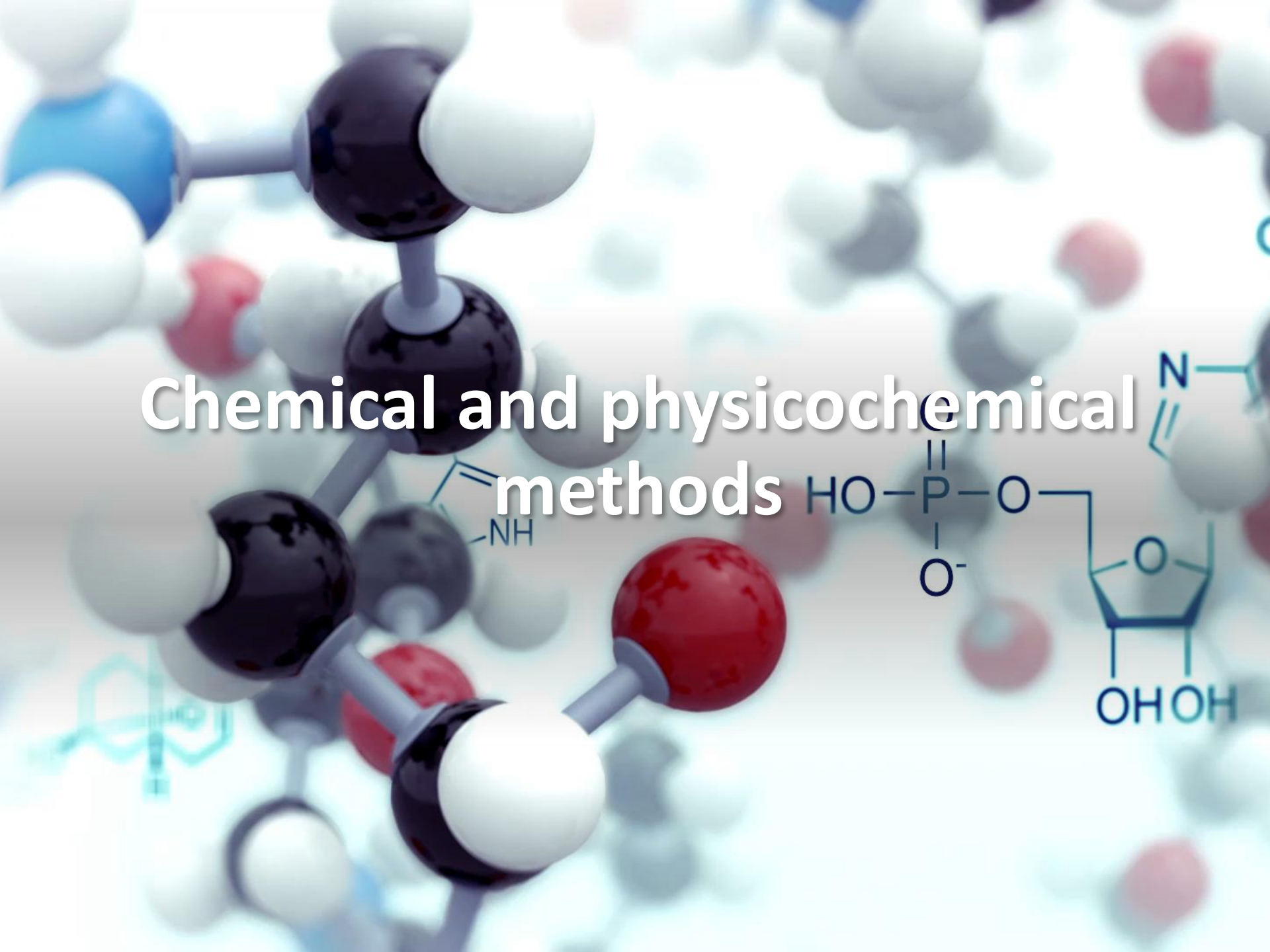
$C_{max} = 5$ mg/ml

$\theta = 50$ s

Therefore when $t = 240$ s:

$$C = 5\left(1 - \exp\left(-\frac{240}{50}\right)\right) \text{ mg/ml} = 4.96 \text{ mg/ml}$$

Chemical and physicochemical methods



Chemical and physicochemical methods

- Detergents
- Enzymes
- Organic solvents
- Osmotic shock
- Subcritical water
- Supercritical water

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Detergents

- Disrupt the membranes cell structure by solubilizing their phospholipids.
- Detergents are classified into cationic, anionic and non-ionic.
- Non-ionic detergents (Triton-X and Tween) are preferred in bioprocessing due to least amount of damage to sensitive biological molecules (proteins and DNA).
- Application: mammalian, bacterial and fungal cells.
- Weakness of this process: Some proteins may denature or precipitate depending on detergent concentration and type.
- Detergent needs to be removed from the product.

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Detergents

Type	Example	Charge	Common Use	Effect on Proteins
Anionic	SDS (Sodium Dodecyl Sulfate), Sodium deoxycholate	Negative	Protein solubilization, SDS-PAGE	Strong denaturant
Cationic	CTAB (Cetyltrimethylammonium bromide), BAC (Benzalkonium chloride)	Positive	Antimicrobial, cell lysis	Strong denaturant, often toxic
Non-ionic	Triton X-100, Tween 20, NP-40	None	Gentle lysis for enzyme/DNA recovery	Mild, preserves protein function

Enyzmes (Lysozyme)

- Lysozyme hydrolyzes the β-1,4-glycosidic bonds in peptidoglycan, lysing bacterial cell walls, especially of Gram-positive bacteria.
- Commonly used together with detergents, osmotic pressure, or mechanical disruption for enhanced efficiency.
- Limitation: High cost and limited effectiveness on Gram-negative bacteria unless combined with EDTA or detergents.
- Application: Mainly for **bacterial cells**.
- Lysozyme needs to be removed from the product

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Organic Solvent

- Organic solvent like acetone, ethanol mainly act on the cell membrane by solubilizing its phospholipids and denaturing proteins.

- Some solvents like toluene can disrupt fungal cell walls.

- Limitation

1. large number of proteins denature and precipitate in the present of organic solvent.
2. need to be subsequently removed from the product where it usually involves additional polishing/ purification.

- Organic solvents are easier to remove (due to volatility) than detergents
- **Modern, eco-friendly alternatives**:

•**Ionic liquids** (e.g., choline chloride–based deep eutectic solvents)

•**Supercritical CO_2**

•**Natural deep eutectic solvents (NADES)** — e.g., choline chloride + glycerol, choline chloride + lactic acid

•**Bio-based solvents** — ethanol, ethyl lactate, limonene

•**Subcritical water (pressurized hot water extraction)**

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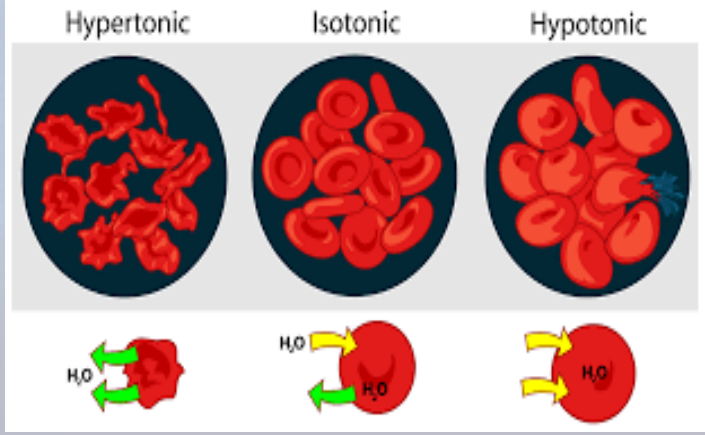
Osmotic Shock

- Osmotic pressure results from a difference in solute concentration across a semi permeable membrane.
- Cell membranes are semi-permeable.
- A sudden transferring a cell from an isotonic medium to hypotonic would cause of a rapid influx of water into the cell.
- This would result in the rapid expansion in cell volume followed by its rupture.
- Ex. Releasing of hemoglobin from red blood cells.
- Application: mammalian, bacterial and fungal cells

https://www.youtube.com/watch?v=A8cI6FkcG4c



- If a cell is placed in a hypertonic solution, there will be a net flow of water out of the cell, and the cell will lose volume. A solution will be hypertonic to a cell if its solute concentration is higher than that inside the cell, and the solutes cannot cross the membrane.
- If a cell is placed in a hypotonic solution, there will be a net flow of water into the cell, and the cell will gain volume. If the solute concentration outside the cell is lower than inside the cell, and the solutes cannot cross the membrane, then that solution is hypotonic to the cell.
- If a cell is placed in an isotonic solution, there will be no net flow of water into or out of the cell, and the cell's volume will remain stable. If the solute concentration outside the cell is the same as inside the cell, and the solutes cannot cross the membrane, then that solution is isotonic to the cell.

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Osmotic Shock

- Osmotic shock is used to removed periplasmic substance (i.e proteins) from cells without physical cell disruption.
- Recombinant, non-recombinant, target proteins are secreted into the periplasmic space.
- If such cells are transferred to hypotonic buffers, the cells imbibe water through osmosis and the volume confined by the cell membrane increase significantly.
- The material present in the periplasmic space is expelled out into the liquid medium.

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New techniques for cell disruption?

- Pulse electric field
- Microwave
- Supercritical extraction
- Subcritical water extraction

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Quiz Time!!!!!!!!!!!!!

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Mind Map

- **Type of products that can be recovered from the cell**
- **List the methods used to disrupt the cell (separate the method into mechanical and non-mechanical)**
- **Brief explanation regarding the mechanism of cell disruption**
- **Find out the efficiency of the cell disruption in terms of the yield, cost, and energy consumption**

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